Test report 14 – Sketches and namespaces

Plan 17.7, step 1: `sketch { … }` with mixed contours, modules as namespaces · cad-syntax, cad-eval, cad-ir, cad-kernel, cad-engine, kernel-smlib (facade `sc_sketch.cpp`), cad-step, cad-session, cad-cli, web · 2026-10-23 · branch worktree-agent-a41d0f69bbe04a86a · commit a2374ac · rustc 1.97.0 (2d8144b78 2026-07-07) · 4 cores

cargo test --workspace	clippy · fmt	Vitest · Playwright	OpenSCAD · STEP	Everyday models
453 / 453	**clean · clean**	**205 · 93 / 93**	**33/33 · 52/52**	**unchanged**
29 of them new/extended	-D warnings	new: S01–S05	compare.py · verify.py (OCCT)	tests/alltag/check.sh

Wasm kernel (Node)
passed
test-node.mjs

1 Task and result

Step 1 of plan 17: **sketches** – a 2D block of segments (lines, circular arcs, Bézier curves, splines), each line of code one segment with its own display, its own handles and its own errors – as a profile for `linear_extrude`, `rotate_extrude`, `loft` and `sweep` (also as a path), exact through the kernel all the way to STEP; and **modules as namespaces** (`teile.schraube(d = 3)`). Draw mode (step 4) and `corner`/`offset`/`stroke` (step 2) are not part of this; `corner()` is reserved (warning).

- Volumes against formulas (section 5): all cases within the integration accuracy of SMLib ($\leq 3.9 \cdot 10^{-6}$ relative; the twist extrusion is a cubic loft approximation as before).
- STEP: arcs as `CIRCLE`, lines as `LINE`, extruded arcs as `CYLINDRICAL_SURFACE`, no approximated edges; OpenCASCADE reads all four new examples with the same volume ($\leq 10^{-6}$).
- OpenSCAD files unchanged: compare.py 33/33; user modules named `line`/`arc` and `spline(points)` outside sketches behave as before (tests).

2 Sketches: language

Decisions (plan 17.8): `sketch` is a built-in 2D module; `sketch {` and `hole {` work without parentheses; segments are ordinary statements that append to the sketch currently being evaluated – also from within modules and loops. Inside a sketch, segment names take precedence over user modules of the same name; outside, everything stays as in OpenSCAD. Other geometry and transformations inside a sketch are an error. Each sketch is **one** node `Profile::Sketch`; the segment statements are stored in `EvalOutput::sketches` so that the cache key does not depend on source positions.

Segment	Forms
`start(p)`	starts a contour; points may be grid points and anchors (z ≠ 0: warning, projected)
`line`	`to = p`, `by = v` (relative), `length = l, angle = a`; `length` alone continues the current direction
`arc`	`to, r [, large, cw]` as in SVG · `center, angle` (degrees, sign = direction) · `center, to` · `through, to` · `to` alone = tangential
`bezier`	`c1, to` (quadratic) · `c1, c2, to` (cubic)
`spline`	points after the current point, `tangent = true | v`, `end_tangent = v`; cubic global interpolation (chord length, averaged knots)
`close()`	line back to the start; if the contour ends at the start, it is closed even without `close()`

`hole { … }`	contours inside are holes (in either direction)

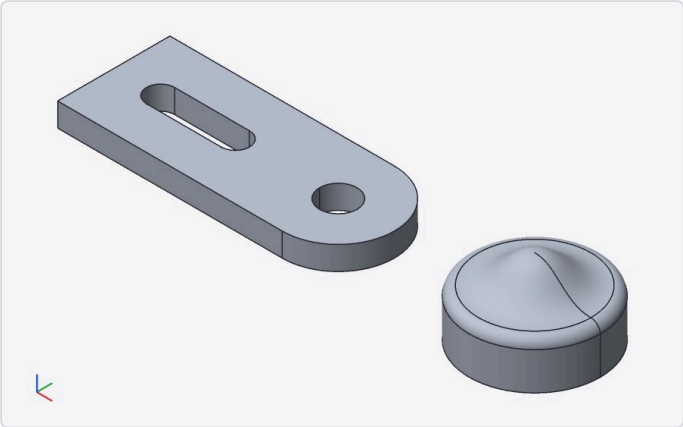
Holes and multiple contours: winding rule *nonzero*; a contour is a boundary where the winding number is ≠ 0 on one side only; islands in holes via nesting. **Sweep:** without `path`, an open sketch (`sketch(closed = false)`) among the children is the path. **Loft:** equal segment count → segment by segment (each segment one face, matched via `start`); otherwise each section becomes a closed B-spline starting at the +x ray from the centroid (as for circles/polygons), so nothing twists.

Examples (`examples/spec/skizze_*.scad`)

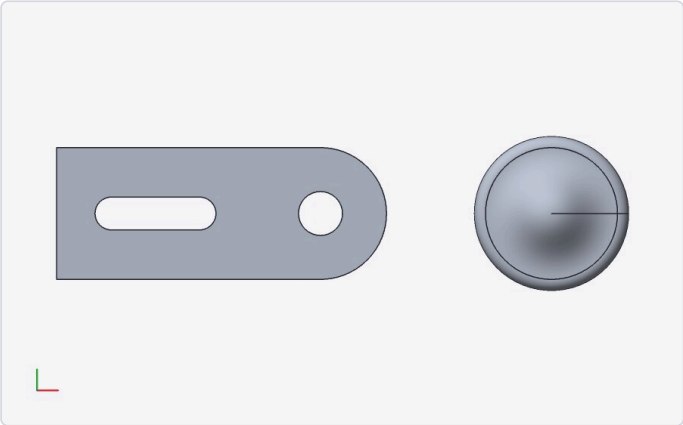
```
// Purpose: sketches with arcs (plan 17.1) — a
bracket as a sketch of lines and arcs with a
// grid inside the sketch, a bore (reverse direction,
winding rule) and a slot
// (`hole { … }`), extruded; next to it a knob from a
sketch with a tangential arc
// and a cubic Bézier curve, revolved about the z
axis.
module halter(l = 60, b = 24, r_loch = 4) sketch {
    grid s {
        x: links = 0, bogen = l - b / 2;
        y: unten = 0, mitte = b / 2, oben = b;
    }
    start(s[links, unten]);
    line(to = s[bogen, unten]);
    arc(to = s[bogen, oben], r = b / 2);        //
half circle at the end
    line(to = s[links, oben]);
    close();
    // bore around the arc center, clockwise: a hole
(winding rule)
    start([l - b / 2 + r_loch, b / 2]);
    arc(to = [l - b / 2 - r_loch, b / 2], r = r_loch,
cw = true);
    arc(to = [l - b / 2 + r_loch, b / 2], r = r_loch,
cw = true);
    // slot: a hole in either direction
    hole {
        start([10, b / 2 - 3]);
        line(to = [26, b / 2 - 3]);
        arc(to = [26, b / 2 + 3], r = 3);
        line(to = [10, b / 2 + 3]);
        arc(to = [10, b / 2 - 3], r = 3);
    }
}

linear_extrude(5) halter();

translate([90, 12, 0]) rotate_extrude() sketch {
    start([0, 0]);
    line(to = [14, 0]);
    line(to = [14, 8]);
    arc(to = [12, 10]);                         //
tangential to the line
    bezier(c1 = [6, 10], c2 = [5, 16], to = [0, 16]);
    close();                                    //
back along the axis
}
```



skizze_profil.scad: bracket (arcs, bore in reverse direction, slot via `hole`), knob (tangential arc, cubic Bézier)

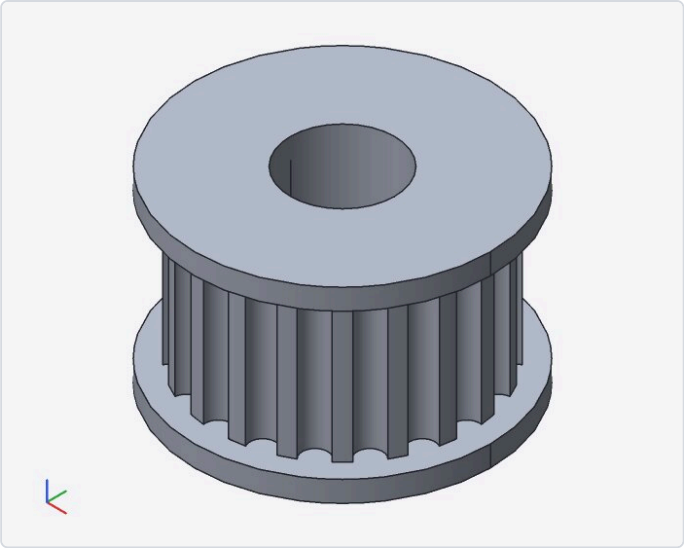


top view

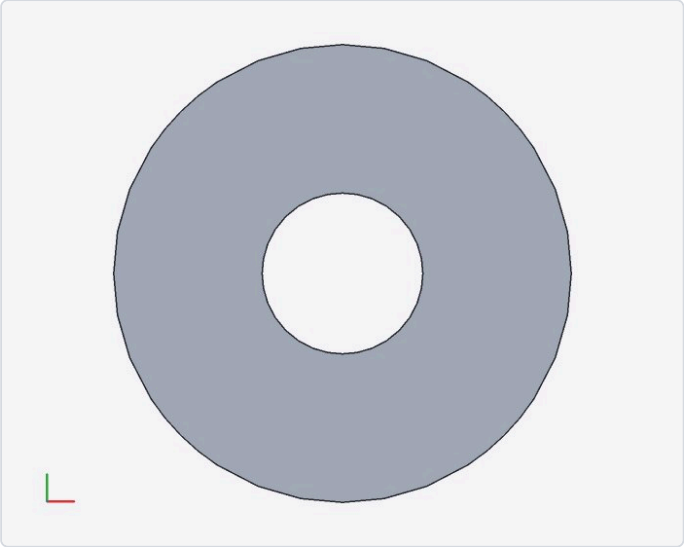
```
// Purpose: timing belt pulley (GT2-like) as a sketch
via a loop (plan 17.1): each tooth gap
// is an arc through a point at the bottom of the
gap, between them an arc around the center on
// the tip circle; the hub bore is a second contour,
clockwise. Flanges as cylinders.
function polar(r, a) = [r * cos(a), r * sin(a)];

module scheibe(z = 20, teilung = 2, h = 7, bohrung =
5) {
    ra = z * teilung / PI / 2 - 0.25;           //
tip circle radius
    t = 0.75;                                   //
depth of the tooth gap
    dw = 0.6 / ra * 180 / PI;                   //
half gap width as an angle
    a = 360 / z;
    translate([0, 0, 1]) linear_extrude(h) sketch {
        start(polar(ra, -dw));
        for (i = [0 : z - 1]) {
            arc(through = polar(ra - t, i * a), to =
polar(ra, i * a + dw));
            arc(center = [0, 0], to = polar(ra, (i +
1) * a - dw));
        }
        start([bohrung / 2, 0]);
        arc(to = [-bohrung / 2, 0], r = bohrung / 2,
cw = true);
        arc(to = [bohrung / 2, 0], r = bohrung / 2,
cw = true);
    }
    for (zb = [0, h + 1]) translate([0, 0, zb])
difference() {
        cylinder(h = 1, r = ra + 1);
        translate([0, 0, -1]) cylinder(h = 3, d =
bohrung);
    }
}

scheibe();
```



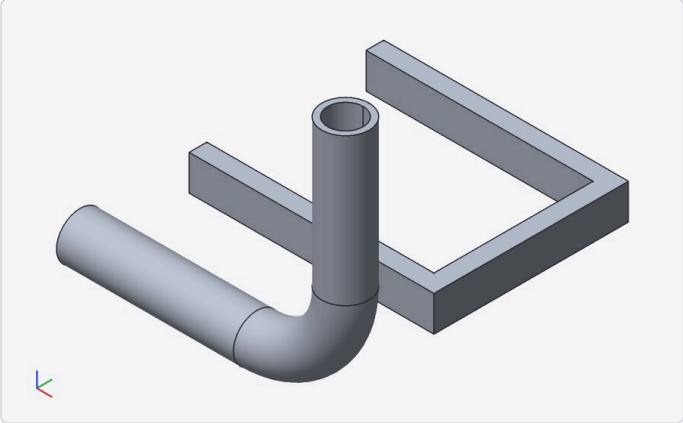
skizze_zahnriemenscheibe.scad: 20 tooth gaps via a loop (arc through a point + arc around the center), hub as a second contour



top view

```
// Purpose: open sketches as a `sweep` path (plan
17.1): a pipe bend of line, tangential
// arc and line (smooth path: one curve chain, the
profile is an annulus) in the xz plane,
// and a frame of a rectangular profile along a path
with corners (miter).
sweep() {
    rotate([90, 0, 0]) sketch(closed = false) {
        start([0, 0]);
        line(to = [30, 0]);
        arc(to = [45, 15]);                    //
tangential: radius 15
        line(by = [0, 25]);
    }
    difference() {
        circle(4);
        circle(3);
    }
}

translate([0, 20, 0]) sweep() {
    sketch(closed = false) {
        start([0, 0]);
        line(to = [40, 0]);
        line(to = [40, 30]);
        line(to = [0, 30]);
    }
    square([3, 6], center = true);
}
```



skizze_sweep.scad: pipe bend (smooth path: one curve chain, annulus as profile), frame along a path with corners (miter)

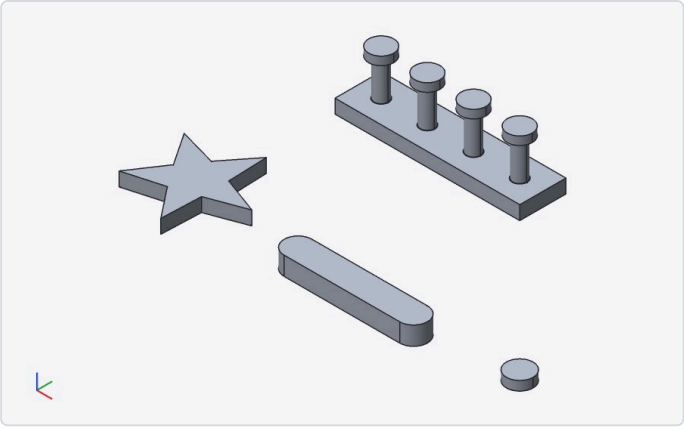
3 Modules as namespaces

`module name { … }` without parentheses; calls `a.b.c(…)`, also named (`k: teile.schraube();`). The first part is looked up lexically, each further part only directly in the body of the previous module. The body is the namespace, with the parameters of the outer module at their default values; it belongs to no instance – the inner module does not see the grids, named instances and anchors of the outer one, and the geometry of the outer module is not created. **Decision on functions/constants:** reachable (`teile.kopf(d)`, `teile.M3`) – libraries bundle helper functions and dimensions with their modules without a separate file; OpenSCAD has no dot after module names (no conflict); a variable of the same name shadows the module. Context: `cad --module teile.platte.lochplatte`, UI breadcrumb "teile › rohr" (S05).

```
// Purpose: modules as namespaces (plan 17.2) — a
library `profile` with sketch modules
// and a parts library `teile` with constants,
functions and submodules; calls from
// outside with a dot (`profile.stern()`,
`teile.schraube(d = 3)`, multi-level
// `teile.platte.lochplatte()`), functions and
constants as `teile.kopf(teile.M3)`.
module profile {
    module stern(n = 5, r1 = 12, r2 = 5) sketch {
        start([r1, 0]);
        for (i = [1 : n]) {
            line(to = [r2 * cos(360 * (i - 0.5) / n),
r2 * sin(360 * (i - 0.5) / n)]);
            line(to = [r1 * cos(360 * i / n), r1 *
sin(360 * i / n)]);
        }
    }
    module langloch(l = 20, b = 6) sketch {
        start([0, -b / 2]);
        line(to = [l, -b / 2]);
        arc(to = [l, b / 2], r = b / 2);
        line(to = [0, b / 2]);
        arc(to = [0, -b / 2], r = b / 2);
    }
}

module teile(raster = 10) {
    M3 = 3.2;                                   // M3
clearance hole
    function kopf(d) = 1.8 * d;                 //
head diameter
    module schraube(d = 3, l = 12) {
        cylinder(h = l, d = d);
        translate([0, 0, l]) cylinder(h = d * 0.6, d
= kopf(d));
    }
    module platte {
        // sees `raster` and `M3` of `teile` (default
values)
        module lochplatte(n = 3) {
            difference() {
                cube([n * raster, raster, 3]);
                for (i = [0 : n - 1])
translate([raster * (i + 0.5), raster / 2, -1])
cylinder(h = 5, d = M3);
            }
        }
    }
}

linear_extrude(3) profile.stern();
translate([25, -3, 0]) linear_extrude(4)
profile.langloch(l = 25);
translate([0, 30, 0]) teile.platte.lochplatte(n = 4);
for (i = [0 : 3]) translate([10 * (i + 0.5), 35, 3])
teile.schraube(d = 3, l = 8);
translate([70, 0, 0]) cylinder(h = 2, d =
teile.kopf(teile.M3));
```



skizze_bibliothek.scad: profile.stern(), profile.langloch(), teile.platte.lochplatte(n = 4), teile.schraube(), disc with d = teile.kopf(teile.M3)

```
examples/spec/skizze_bibliothek.scad: 0 Fehler, 0
Warnungen
Module:
  profile()  skizze_bibliothek.scad:5–20
  profile.stern(n = 5: Zahl, r1 = 12: Zahl, r2 = 5:
Zahl)  skizze_bibliothek.scad:6–12
  profile.langloch(l = 20: Zahl, b = 6: Zahl)
skizze_bibliothek.scad:13–19
  teile(raster = 10: Zahl)
skizze_bibliothek.scad:22–38
  teile.schraube(d = 3: Zahl, l = 12: Zahl)
skizze_bibliothek.scad:25–28
  teile.platte()  skizze_bibliothek.scad:29–37
  teile.platte.lochplatte(n = 3: Zahl)
skizze_bibliothek.scad:31–36
Funktionen:
  teile.kopf(d) = 1.8 * d  skizze_bibliothek.scad:24
Farben (Filament, Farbe, Volumen, Masse PLA 1,24
g/cm³):
```

```
examples/spec/skizze_bibliothek.scad: 0 Fehler, 0
Warnungen
```

4 Kernel integration

- `cad_kernel::Segment::{Line, Arc, Nurbs}` (`curve2.rs`): NURBS form, affine mapping (arcs stay arcs under similarity transforms, otherwise exact rational elliptical arcs), sampling, global interpolation, joining into one B-spline

(Bézier decomposition, degree elevation); `Loop::Composite`, `PlacedPath`, `Kernel::sweep_path`.

- Facade `smlib/facade/sc_sketch.cpp`, included at the end of `sc_facade.cpp` (uses its internal helpers; compiled on its own it is empty). Loop kind 3: lines → `SmLine`, arcs → rational quadratic B-splines (`SM_CF_CIRCULAR_ARC`), Bézier/splines → B-splines; joints welded exactly. Changes to existing functions: one field in `Loop`, one branch each in `parse_loops`/`loop_curves`, the condition in the twist branch of `sc_extrude`, `loft_segmentwise` in `sc_loft`. New: `sc_sketch_sweep` (smooth path: curve sweep with profile reference point from patch 0003; corners: miter).
- STEP: SMLib stores extruded arcs as `SmSurfOfExtrusion` of the exact B-spline; on such surfaces (without p-curves) OpenCASCADE computed volumes that were wrong by up to $2.5 \cdot 10^{-3}$. The writer therefore writes an extrusion of a circular arc along its axis as `CYLINDRICAL_SURFACE` (orientation adjusted) – after that OCCT agrees to 10^{-9} or to the kernel accuracy, respectively.

Model	Faces	CIRCLE	LINE	B-spline curves	Surface types	approximated
skizze_profil	16	14	21	3	cylinder 5, plane 8, revolution 3	0 / 0
skizze_zahnriemenscheibe	47	86	44	0	cylinder 43, plane 4	0 / 0
skizze_sweep	18	10	24	4	nurbs 5, plane 12, revolution 1	0 / 0
rechteck_halbkreis	6	2	10	0	cylinder 1, plane 5	0 / 0
torus	2	4	0	0	revolution 2	0 / 0

OpenCASCADE (`tests/step/verify.py`) for the new examples (columns: bodies, faces, V kernel, V OCCT, dV, dA, dBox):

```
skizze_bibliothek            ok        8    51         2788.8         2788.8 8.685e-07 4.080e-06 3.621e-17
{'cylinder': 15, 'plane': 36}
skizze_profil                ok        2    16        12823.7        12823.7 1.091e-06 1.098e-06 9.184e-10
{'cylinder': 5, 'plane': 8, 'revolution': 3}
skizze_sweep                 ok        2    18        3707.67        3707.67 1.287e-06 7.266e-06 1.161e-09
{'bspline': 5, 'plane': 12, 'revolution': 1}
skizze_zahnriemenscheibe     ok        1    47        860.299        860.299 3.972e-07 4.166e-05 1.611e-16
{'cylinder': 43, 'plane': 4}
```

5 Measurements: volumes against formulas

One small file per case (`reports/tmp/report14/cases/`), `cad measure --volume` with the release build (time: whole CLI call without daemon). The geometry is exact; the deviation is that of the volume integration in SMLib.

Case	Kernel mm³	Formula mm³	rel. dev.	ms
Rectangle 40 × 20 + half circle r 20 (end point + radius), h 5	7,141.5969	7,141.5927	$5.9 \cdot 10^{-7}$	59
same with tangential arc	7,141.5969	7,141.5927	$5.9 \cdot 10^{-7}$	46
same with center + angle	7,141.5969	7,141.5927	$5.9 \cdot 10^{-7}$	50
rotate_extrude of a half circle: sphere r 7	1,436.7565	1,436.7550	$1.0 \cdot 10^{-6}$	81
rotate_extrude of two arcs: torus R 20, r 4	6,316.5572	6,316.5468	$1.6 \cdot 10^{-6}$	114
Square 20 minus circle r 5 (reverse direction, nonzero)	642.9202	642.9204	$2.6 \cdot 10^{-7}$	70
same with `hole { … }`	642.9202	642.9204	$2.6 \cdot 10^{-7}$	71
Square 4 + quadratic Bézier cap (parabola 2/3 · b · h)	18.6666	18.6667	$1.0 \cdot 10^{-6}$	49
Star of 12 lines via a loop in a sketch module	450.0000	450.0000	$1.3 \cdot 10^{-16}$	51
sweep of a circle r 2 along line–arc–line (Pappus)	637.2176	637.2151	$3.9 \cdot 10^{-6}$	104
sweep of a square 2 along a corner (miter)	140.0000	140.0000	$4.1 \cdot 10^{-16}$	68

linear_extrude(twist = 90) of a slot (volume without twist)	882.7768	882.7433	$3.8 \cdot 10^{-5}$	72

twist: the extrusion with twist is (as before for all profiles) a cubic loft approximation through sections every 7.5°.

6 Errors per segment line

```
linear_extrude(2) sketch {
    start([0, 0]);
    line(to = [30, 0]);
    arc(to = [30, 20], r = 5);
    line(to = [0, 20]);
}
linear_extrude(2) translate([40, 0]) sketch {
    start([0, 0]);
    line(to = [10, 10]);
    line(to = [10, 0]);
    line(to = [0, 10]);
    close();
}
```

```
cases/fehler.scad:4:5: Fehler (Auswertung): unmöglicher Bogen: Radius 5 kleiner als die halbe Sehne 10
cases/fehler.scad:5:5: Fehler (Auswertung): Kontur nicht geschlossen: endet bei [0, 20] statt [0, 0]; `close()` fehlt
cases/fehler.scad:11:5: Fehler (Auswertung): Selbstüberschneidung der Kontur mit Segment in Zeile 9
cases/fehler.scad:7:1: Fehler (Kernel): linear_extrude gescheitert: extrude did not produce a closed solid (self-intersecting or open profile?)
cases/fehler.scad: 4 Fehler, 0 Warnungen
```

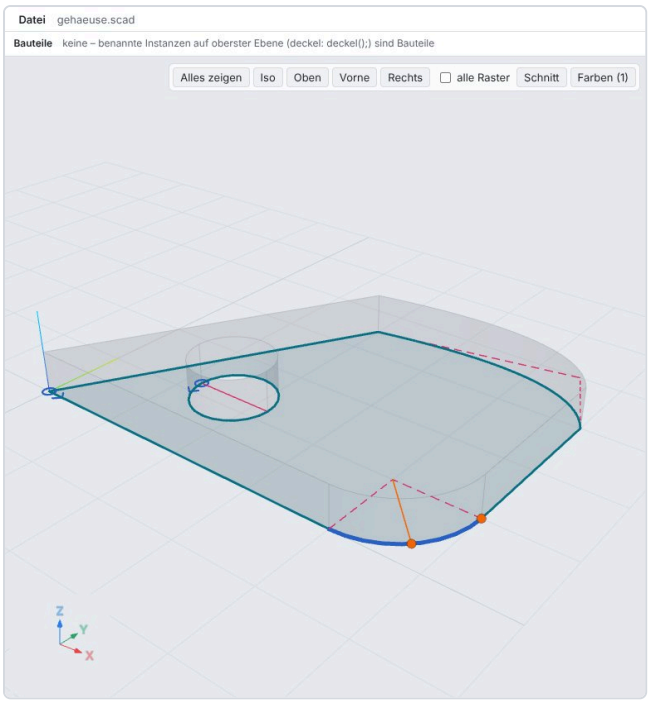
Checked are: contour not closed (the sketch stays visible with a red dashed closing line), self-intersection (reported at the later line, with a reference), impossible arc (radius < half chord, three collinear points, tangential along the tangent, angle 0 or > 360°), contour without area, segment before `start`/after `close()`. A self-intersecting contour is additionally reported by the kernel at the extrusion.

7 UI: display and handles

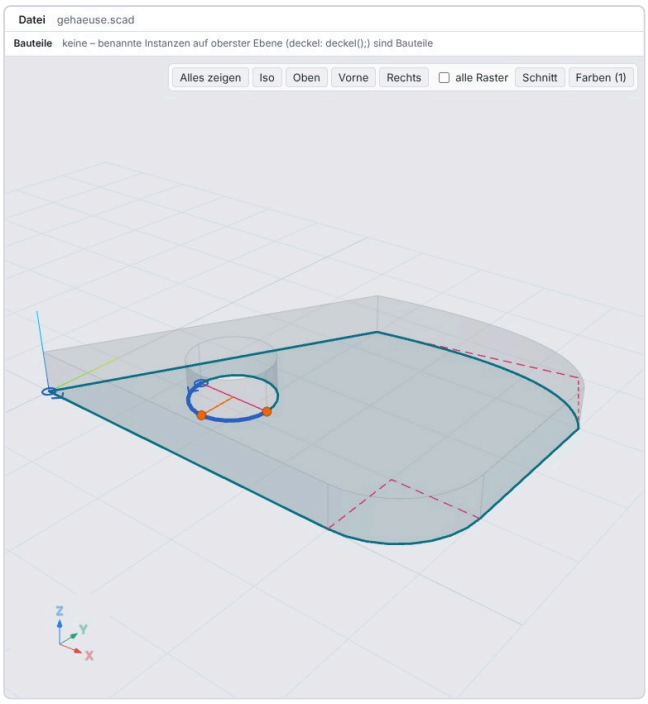
When the cursor is in a sketch (segment line, loop, `hole`, call of a sketch module), the view draws it in its plane: mesh as an outline with a light fill (visible through the body), segments as lines, the segment of the cursor line thick in the accent color, errors red, control polygons and arc centers dashed, the start as a ring with a direction arrow, the end of open contours as a square. Handles: end/start points (literal or grid snap; `line(by = …)` relative, without snap), arc radius, center, through point, Bézier control points, spline points. Checks via the UI state (`window.__cad`) and pixels of the layers.

S01 Sketch under the cursor

Cursor on a segment line: the sketch is drawn in its plane – outline with a light fill (mesh from the engine, mode "profile", visible through the body), the segments as lines, the segment of the line thick in the accent color; the start point of each contour as a ring with a direction arrow, arc centers as dashed helper lines.



Cursor on arc(...) (line 4)



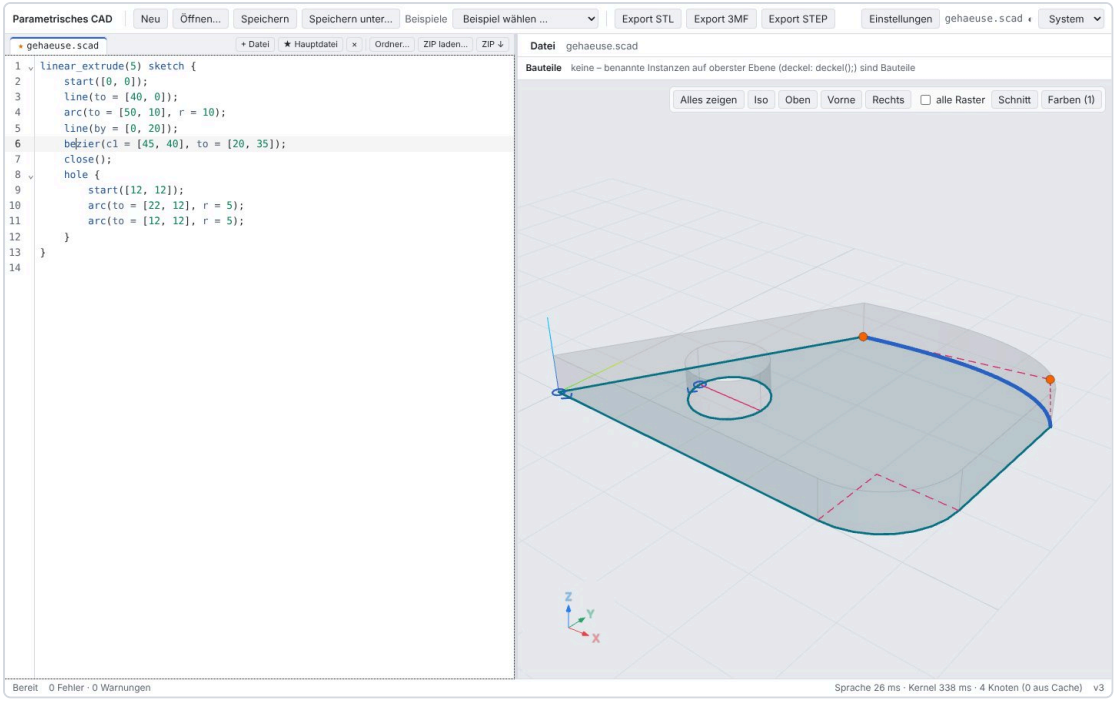
Cursor in the hole block (line 10)

```
linear_extrude(5) sketch {
    start([0, 0]);
    line(to = [40, 0]);
    arc(to = [50, 10], r = 10);
    line(by = [0, 20]);
    bezier(c1 = [45, 40], to = [20, 35]);
    close();
    hole {
        start([12, 12]);
        arc(to = [22, 12], r = 5);
        arc(to = [12, 12], r = 5);
    }
}
```

- active segment: arc (line 4), 1019 pixels
- start markers: 2
- line 10: arc in the hole active

S02 Bézier segment

Quadratic Bézier curve: control polygon (start – c1 – end) dashed as a helper line, handles at the end point and at the control point.



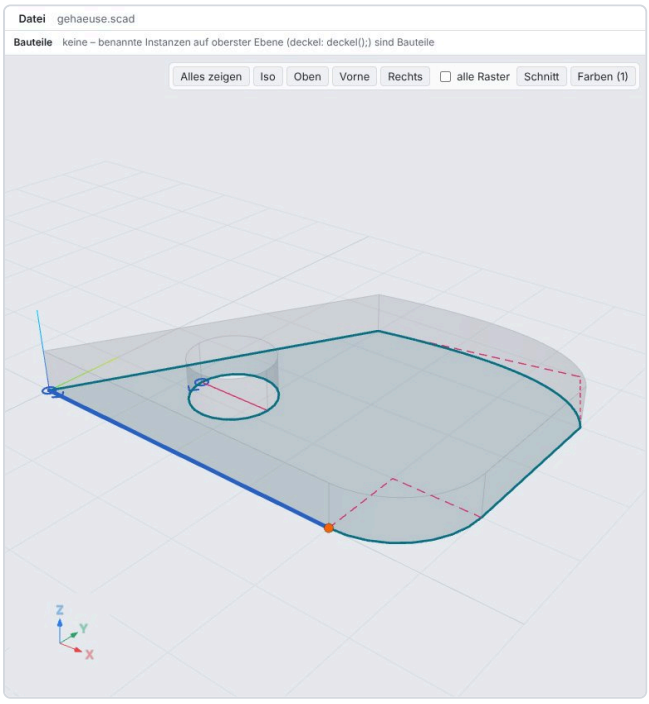
Cursor on bezier(c1 = ..., to = ...)

```
bezier(c1 = [45, 40], to = [20, 35]);
```

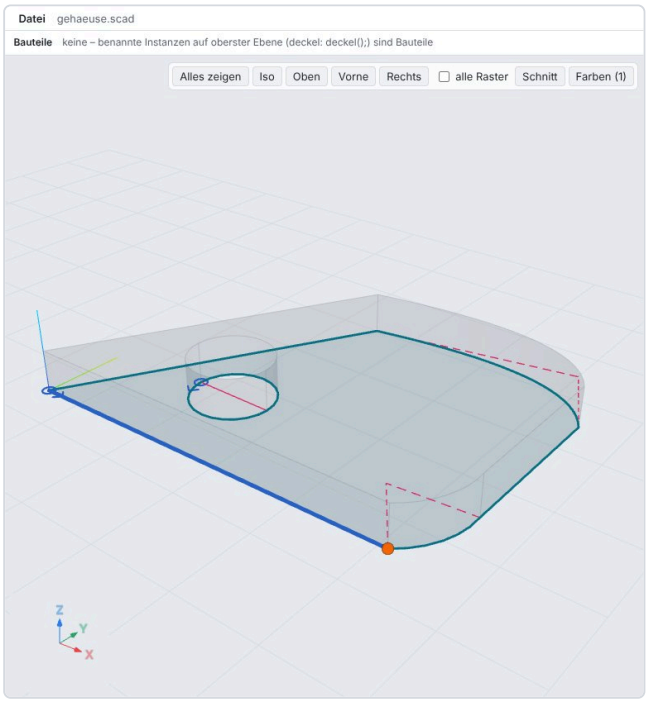
- helper lines: 4
- pixels of the helper lines: 735
- 2 point handles (to, c1)

S03 Dragging an end point

The point handle at the end of the line dragged with the mouse (Alt: free, grid 1 mm): the literal in the line changes, the sketch follows live; the following arc starts at the new point.



before: line(to = [40, 0])



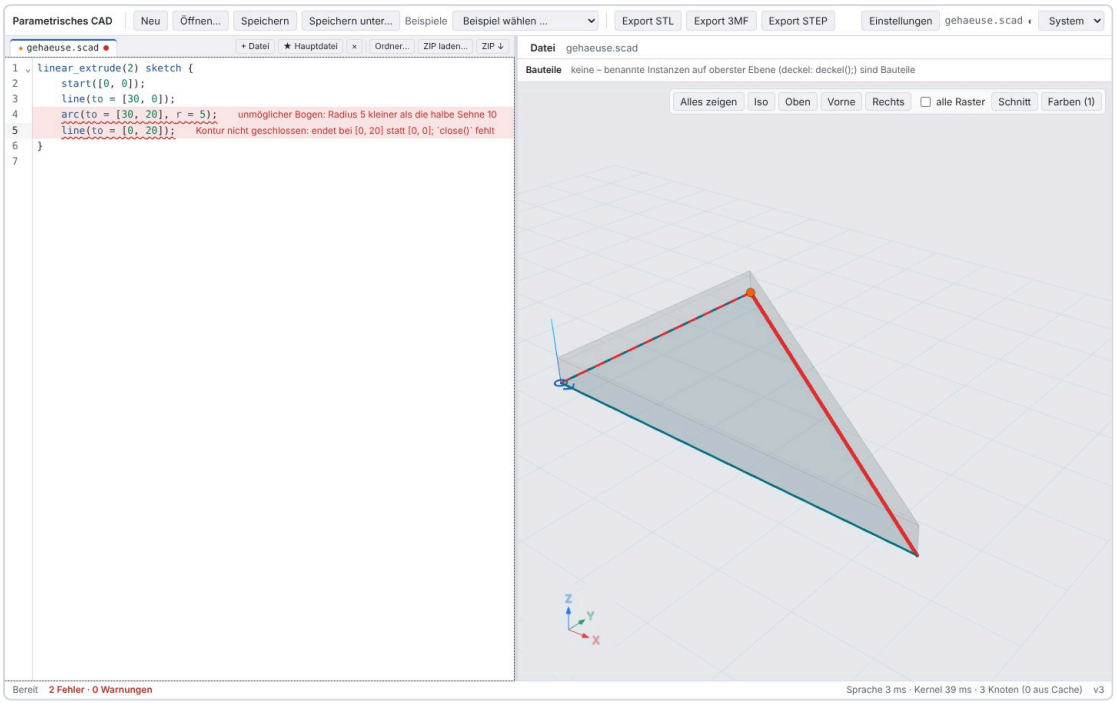
after: line(to = [46, 1])

```
// before
line(to = [40, 0]);
// after dragging (Alt, free)
line(to = [46, 1]);
```

- literal: [40, 0] → [46, 1]
- arc line unchanged

S04 Errors per segment line

An arc with a radius smaller than half the chord and a contour without close(): both errors are shown at their segment line (red, message at the end of the line); the sketch stays visible with the missing closing line (red dashed).



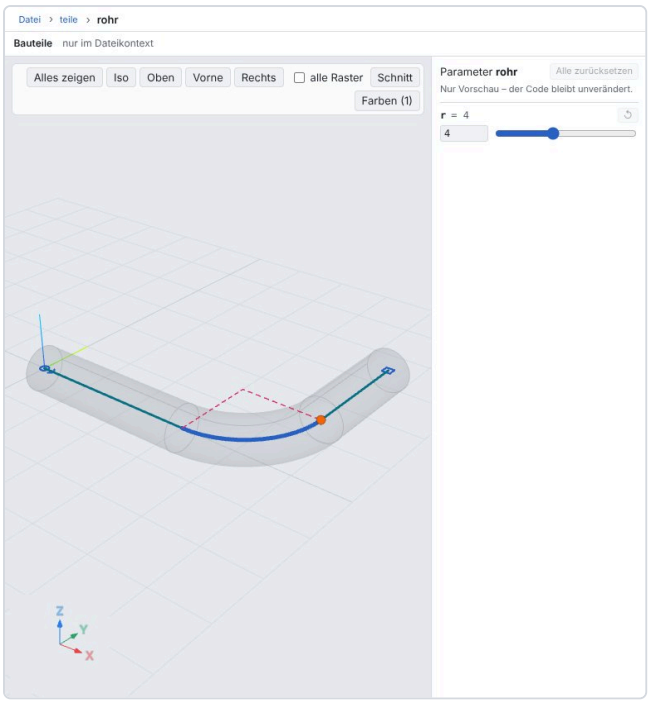
impossible arc, contour not closed

```
linear_extrude(2) sketch {
    start([0, 0]);
    line(to = [30, 0]);
    arc(to = [30, 20], r = 5);
    line(to = [0, 20]);
}
```

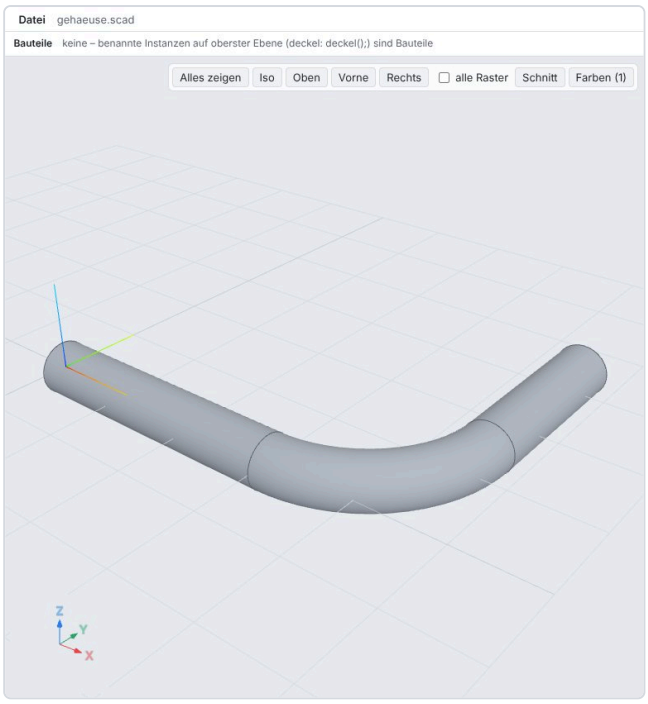
- 4: unmöglicher Bogen: Radius 5 kleiner als die halbe Sehne 10 (impossible arc: radius 5 smaller than half the chord 10)
- 5: Kontur nicht geschlossen: endet bei [0, 20] statt [0, 0]; `close()` fehlt (contour not closed: ends at [0, 20] instead of [0, 0]; `close()` missing)

S05 Sweep path and namespace

An open sketch (closed = false) is the path of the sweep: without fill, start as a ring, end as a square; the circle runs along line – tangential arc – line (one curve chain). The module rohr is in the namespace teile: the breadcrumb shows "teile › rohr"; from outside it is called as teile.rohr(r = 3).



Context teile › rohr, cursor on the arc of the path



Call teile.rohr(r = 3) in the file

```
module teile {
    module rohr(r = 4) {
        sweep() {
            sketch(closed = false) {
                start([0, 0]);
                line(to = [30, 0]);
                arc(to = [45, 15]);
                line(by = [0, 20]);
            }
            circle(r);
        }
    }
}
teile.rohr(r = 3);
```

- breadcrumb: teile › rohr
- open contour: 1, end marker drawn
- call in the file fully displayed

8 Test runs

Run	Result
`cargo test --workspace --no-fail-fast`	453 passed, 0 failed
`cargo clippy --workspace --all-targets -- -D warnings` · `cargo fmt --check`	clean · clean
`pnpm typecheck` · `pnpm test` (Vitest)	clean · 205 passed
`pnpm e2e` (Playwright, Chromium)	93 / 93 · second run; first run 92/93: B10 (snap of a zone center, report 3) failed once, passed on its own and in the second run
`tests/openscad-compat/compare.py --eval`	33/33
`tests/step/verify.py` (OpenCASCADE)	52/52 passed
`tests/alltag/check.sh`	19 models, 0 with problems, 87 s, exports in /tmp/alltag-ocn57h6z (deleted)

`node crates/kernel-smlib/js/test-node.mjs`	passed

New or extended Rust tests (29):

`sketch_and_namespaces` **ok** · `bezier_and_spline_segments` **ok** · `loft_between_sketches` **ok** · `sketch_alone_is_2d_and_open_sketch_is_a_path` **ok** · `holes_by_winding_and_hole_block` **ok** · `revolve_arc_profiles_sphere_and_torus` **ok** · `sketch_modules_loops_and_twist` **ok** · `rectangle_with_half_circle` **ok** · `step_has_exact_lines_and_circles` **ok** · `sweep_along_open_sketch_path` **ok** · `arc_forms` **ok** · `named_sketch_module_and_qualified_profiles` **ok** · `namespace_visibility_and_errors` **ok** · `handles_of_segments_and_dragging` **ok** · `openscad_files_are_unaffected` **ok** · `errors_per_segment_line` **ok** · `qualified_module_calls` **ok** · `loops_variables_grids_and_modules_in_a_sketch` **ok** · `several_contours_holes_and_open_paths` **ok** · `qualified_context_module` **ok** · `segments_of_all_kinds` **ok** · `curve2::tests::arc_nurbs_lies_on_circle` **ok** · `curve2::tests::join_keeps_the_geometry` **ok** · `curve2::tests::mapped_arcs` **ok** · `curve2::tests::interpolation_hits_points_and_tangents` **ok** · `tests::module_without_parentheses_and_qualified_calls` **ok** · `tests::sketch_block_without_arguments` **ok** · `spec_models_stage1` **ok** · `sweep_along_path` **ok**

Vitest: `web/src/logic/sketch.test.ts` (sketch under the cursor, helper lines, segment handles, module with path); Lezer grammar for `sketch {`, modules without parentheses and `a.b()`. CLI: `sketch_and_namespaces` (volume, function in a namespace, `outline`, `--module teile.platte`, errors per line).

9 Limitations

- `corner`, `fillet2d`, `offset`, `stroke` are missing (step 2); no transformations inside a sketch (a module called in a sketch draws in sketch coordinates).
- Self-intersection is checked on sampled polylines (tolerance relative to the sketch size); intersections between contours are allowed (nonzero); the nesting of holes and islands assumes non-intersecting contours.
- Sweep along a sketch with corners only as a miter and with the profile centered on the origin; closed paths are not supported.
- Loft with different segment counts joins each section into one curve (C0 at the joints): the faces no longer follow the segments.
- STEP: only extrusions of circular arcs are written as cylinders; surfaces of revolution of arcs remain `SURFACE_OF_REVOLUTION` of the exact curve (sphere/torus not recognized; OCCT reads them correctly).
- `cad render --line N` of a segment line highlights the whole sketch; highlighting of the single segment exists only in the UI.
- Namespaces: constants are evaluated in a new namespace on each access (no caching); `--module name` without a path takes the first module of that name.